NETFLIX AFTER HOURS

PRODUCT REQUIREMENT DOCUMENT (PRD) — COMPREHENSIVE

Version: 2.0-FINAL
Date: Q2 2027 Launch
Owner: Product & UI/UX Strategy

1. EXECUTIVE SUMMARY & VISION

Netflix After Hours is a strategic product initiative designed to transform late-night passive viewing into an intelligent, circadian-friendly experience. Between 11:00 PM and 3:00 AM, up to 25% of streaming bandwidth runs without active user engagement as viewers fall asleep. This leads to redundant data delivery costs, accidental season spoilers, and corrupted viewing history. *After Hours* introduces auto-dimming display profiles, adaptive audio decay, intelligent sleep-point bookmarking, and ecosystem wearable integration to turn bedtime habits into a seamless retention and monetization engine.

2. STRATEGIC OBJECTIVES & SUCCESS METRICS

Metric Type	Key Performance Indicator (KPI)	Target Objective
Primary / Financial	Bandwidth Cost Reduction ($/TB)	Reduce late-night non-interactive data delivery by 12% globally.
User Experience	Sleep Resume Accuracy Rate	>85% of users resuming via Sleep Card complete without manual rewind.
Monetization	Premium Tier Upsell Rate	Drive 1.8% upgrade conversion via Wearable Sync & Audio-Only mode.
Ad Protection	Ad Viewability Integrity	Zero unviewed video ads served during active screen-dimmed states.

3. PLAN TIERING & PACKAGING MATRIX

Feature Capability	Standard with Ads	Standard (Ad-Free)	Premium (4K + Spatial)
Manual Sleep Timers (15 / 30 / 45 / 60m / End of Ep)	✓	✓	✓
Sleep-State Auto-Bookmarking & Recovery	✓	✓	✓
OLED Circadian Display Dimming Curve	✓	✓	✓
Audio-Only Background Streaming Mode	—	✓	✓
Smart Wearable Auto-Sleep Detection	—	—	✓
Spatial Audio Binaural Sleep Soundscape	—	—	✓

4. DETAILED USER STORIES & ACCEPTANCE CRITERIA

US-01: Manual Sleep Timer Setup

"As a late-night viewer, I want to set a sleep timer directly from player controls so playback stops automatically without keeping my TV running all night."

- Moon icon accessible in player control bar on TV, Mobile, and Web.
- Menu options: 15m, 30m, 45m, 60m, End of Episode, and Custom duration.
- Selecting a timer shows a subtle 3-second confirmation toast ("Sleep Timer set for 30m").

US-02: Display Dimming & Audio Fading Decay

"As a sleeping viewer, I want video to turn black and audio to fade out gradually before stopping, so sudden silence does not wake me up."

- 10m before expiry, player lowers screen brightness exponentially to #000000.
- 5m before expiry, audio cuts treble (>8kHz) and bass (<80Hz) while volume drops to 15%.
- On Standard with Ads, scheduled ad breaks during dimming are deferred until active playback resumes.

US-03: Sleep-State Auto-Bookmarking & Recovery

"As a viewer who fell asleep, I want Netflix to remember where my video started fading so I can resume easily without guessing timestamps."

- System sets a Decay Bookmark at the exact timestamp dimming initiated.
- A Sleep Recovery Card displays at the top of the Netflix Home feed next day.
- Tapping "Resume from Sleep Point" launches playback at exact decay timestamp.

5. TECHNICAL SPECIFICATIONS & ARCHITECTURAL FLOW

Architecture Rule: The dimming transition operates on a 10-minute decay curve to true black (#000000). Audio filters drop high frequencies and low sub-bass 5 minutes prior to session shutdown.

```
// Sleep State Event Sync Payload (Fired to Sync Engine) { "account_id": "usr_99201481",
"profile_id": "prf_88201", "device_type": "smart_tv_android", "decay_bookmark_timestamp_ms":
1420500, "trigger_mechanism": "wearable_auto_healthkit", "tier_eligibility": "premium",
"ad_deferral_active": true }
```

6. UI/UX ERGONOMICS & CIRCADIAN DESIGN

- **Circadian Spectrum Protection:** All After Hours UI overlays strictly enforce a dark amber palette (#FF4500 accents over #000000 OLED true-black) to eliminate blue-light wavelength disruption during bedtime use.
- **Low-Motor-Precision Touch Architecture:** Control targets in After Hours mode expand by 30% (minimum touch area 56x56dp), enabling effortless touch manipulation in complete darkness without direct eye focus.

7. COMPLETE EDGE CASES & RISK MITIGATION MATRIX

Edge Case Scenario	Potential Risk / Impact	Mitigation Strategy & System Behavior
Multiple Viewers in Room	Wearable triggers sleep mode while co-viewers remain awake.	TV overlay displays a non-blocking 15-second toast: *"Sleep detected via Apple Watch. Press any button to cancel."*
Mid-Episode Ad Break (Ad Tier)	Ad plays during screen dimming, causing advertiser ROI degradation.	Ad breaks pause during fade curve. Deferred ad units return to server queue for active resume playback.
Wearable Sync Disconnection	Bluetooth/Health API connection drops mid-stream.	System fails over automatically to standard 60-minute inactivity timer check.
Background Audio Termination	Mobile OS kills background audio process to save battery.	Request native Foreground Media Service permission with persistent player notification widget on iOS/Android.
Network Sync Failure on Shutdown	Sleep timestamp fails to write to cloud database due to local outage.	Cache local timestamp in encrypted local device storage; queue payload sync on next successful handshake.